

Critical Ideals of Ideal Schauder Systems: A Verified Obstruction and Operator-Theoretic Reductions

Research note prepared from an automatic-solver packet

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Abstract

Kwela and Swaczyna ask whether every analytic P -ideal is the critical ideal of an ideal Schauder system in a Banach space. This note gives the strongest conclusions obtained in a renewed attempt at that problem. First, it repairs and proves the packet's unconditional-support theorem: every ideal defined by convergence of positive weighted subseries along a 1-unconditional basis is non-pathological. This is not a new characterization; it is a transparent special case of the Banach-representability theorem of Borodulin-Nadzieja-Farkas-Plebanek.

For a general ideal Schauder system, the note then rewrites the problem in terms of the residual projections $R_n = I - S_n$. Combining this reformulation with a Katetov characterization of non-pathological analytic P -ideals gives an exact statistical strong-operator selection criterion and an equivalent regular-matrix averaging criterion. Two usable sufficient conditions follow. In particular, if every positive set contains a positive subset on which $(\|R_n\|)$ is bounded, then the critical ideal is non-pathological. Using the filter-dependent uniform boundedness principle of De Bondt and Vernaev, this holds whenever the dual filter is rapid⁺. Finally, a theorem of Kadets and Manskova yields a necessary reciprocal-norm divergence condition on every positive set.

No complete proof or counterexample to the source question was located or obtained. The exact remaining obstruction is a statistical strong-operator selection problem for an unbounded nested family of projections.

1 The source question and the status of this note

Let X be a Banach space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and let $\hat{a} = (a_n, a_n^*)_{n \in \mathbb{N}} \subseteq X \times X^*$ be biorthogonal: $a_n^*(a_m) = \delta_{nm}$. Put

$$S_n x = \sum_{i=0}^n a_i^*(x) a_i, \quad R_n x = x - S_n x.$$

For an ideal $\mathcal{I}$ on $\mathbb{N}$, the system is an $\mathcal{I}$ -Schauder system if $R_n x \rightarrow 0$ along the dual filter $\mathcal{I}^*$ for every $x \in X$. The least ideal with this property is its *critical ideal*, denoted $\mathcal{CR}(X, \hat{a})$. Equivalently,

$$\mathcal{CR}(X, \hat{a}) = \text{Ideal}(\text{Fin} \cup \{B(x, \varepsilon) : x \in X, \varepsilon > 0\}), \quad B(x, \varepsilon) = \{n : \|R_n x\| \geq \varepsilon\}. \quad (1)$$

Here $\text{Ideal}(\mathcal{A})$ denotes the ideal generated by the family $\mathcal{A}$.

Kwela and Swaczyna proved that every non-pathological analytic P -ideal is a critical ideal and asked the following question [7, Question 11.2]:

Is every analytic P -ideal of the form $\mathcal{CR}(X, \hat{a})$ for some Banach space X and some biorthogonal system $\hat{a}$ with continuous coordinate functionals?

The published version of their paper still contains the question. The literature search carried out for this note did not locate a subsequent paper or preprint resolving it. Thus the statements below are presented as partial progress and reductions, not as a claimed solution of Question 11.2.

2 Preliminaries on analytic P -ideals

A function $\phi : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ is a *submeasure* if $\phi(\emptyset) = 0$, $\phi(\{n\}) < \infty$, it is monotone, and $\phi(A \cup C) \leq \phi(A) + \phi(C)$. It is lower semicontinuous if

$$\phi(A) = \sup_m \phi(A \cap m).$$

Set

$$\text{Exh}(\phi) = \{A \subseteq \mathbb{N} : \lim_{m \rightarrow \infty} \phi(A \setminus m) = 0\}.$$

Solecki's theorem says that the analytic P -ideals are exactly the ideals $\text{Exh}(\phi)$ for lower semicontinuous submeasures ϕ [8].

A lower semicontinuous submeasure ϕ is *non-pathological* if

$$\phi(A) = \sup\{\mu(A) : \mu \text{ is a finitely supported measure and } \mu \leq \phi\} \quad (2)$$

for every $A \subseteq \mathbb{N}$. An analytic P -ideal is non-pathological if it is $\text{Exh}(\phi)$ for such a ϕ .

We shall use the Katetov order. For ideals $\mathcal{I}$ on M and $\mathcal{J}$ on N , write $\mathcal{I} \leq_K \mathcal{J}$ if there is $f : N \rightarrow M$ such that $f^{-1}[C] \in \mathcal{J}$ for every $C \in \mathcal{I}$. Let

$$\mathcal{I}_d = \left\{ A \subseteq \mathbb{N} : \frac{|A \cap m|}{m} \rightarrow 0 \right\}$$

be the asymptotic-density-zero ideal. The following characterization is the main descriptive-set-theoretic input used later.

Theorem 2.1 (Hrušák; see Filipów–Tryba). *For an analytic P -ideal $\mathcal{I}$, the following are equivalent:*

- (i) $\mathcal{I}$ is non-pathological;
- (ii) $\mathcal{I} \upharpoonright A \leq_K \mathcal{I}_d$ for every $A \notin \mathcal{I}$.

Moreover, an arbitrary ideal is an intersection of regular matrix summability ideals if and only if it satisfies condition (ii).

The first assertion is recorded as Theorem 8.4 and the matrix assertion as Theorem 9.1 in [5].

3 The unconditional support theorem

The automatic packet proposed the theorem below. Its conclusion is correct, but four details require care: the subadditivity split must be disjoint; $\phi(F)$ must first be identified with the norm of the full positive sum; the Cauchy argument should use ordered tails; and the positive norming functional must be justified. We include a complete proof.

Theorem 3.1 (Unconditional support ideals are non-pathological). *Let X have a 1-unconditional Schauder basis $(u_n)_{n \in \mathbb{N}}$, and let $c_n > 0$. Define*

$$\mathcal{J}_{X,c} = \left\{ A \subseteq \mathbb{N} : \sum_{n \in A} c_n u_n \text{ converges in } X \right\}$$

and

$$\phi(A) = \sup \left\{ \left\| \sum_{n \in F} c_n u_n \right\| : F \in [A]^{<\omega} \right\}. \quad (3)$$

Then ϕ is a non-pathological lower semicontinuous submeasure and

$$\mathcal{J}_{X,c} = \text{Exh}(\phi).$$

Proof. Monotonicity and $\phi(\emptyset) = 0$ are immediate, and $\phi(\{n\}) = c_n \|u_n\| < \infty$. If $F \subseteq A \cup C$ is finite, use the disjoint decomposition

$$F = (F \cap A) \dot{\cup} (F \setminus A), \quad F \setminus A \subseteq C.$$

The triangle inequality gives

$$\left\| \sum_{n \in F} c_n u_n \right\| \leq \phi(A) + \phi(C).$$

Taking the supremum over F proves subadditivity. Also

$$\phi(A) = \sup_m \phi(A \cap m),$$

so ϕ is lower semicontinuous.

Suppose first that $\phi(A \setminus m) \rightarrow 0$. If $p > q \geq m$, then

$$\left\| \sum_{\substack{n \in A \\ q \leq n < p}} c_n u_n \right\| \leq \phi(A \setminus m).$$

Thus the ordered partial sums of the subseries form a Cauchy sequence, and the subseries converges.

Conversely, suppose that the subseries converges and let

$$z_m = \sum_{\substack{n \in A \\ n \geq m}} c_n u_n.$$

A 1-unconditional basis has the lattice-monotonicity property: if $0 \leq y \leq z$ coordinatewise, then $\|y\| \leq \|z\|$. Hence, for every finite $F \subseteq A \setminus m$,

$$\left\| \sum_{n \in F} c_n u_n \right\| \leq \|z_m\|.$$

Therefore $\phi(A \setminus m) \leq \|z_m\| \rightarrow 0$, proving $\mathcal{J}_{X,c} = \text{Exh}(\phi)$.

It remains to prove non-pathology. First note that for finite F ,

$$\phi(F) = \|x_F\|, \quad x_F = \sum_{n \in F} c_n u_n. \quad (4)$$

Indeed, each positive subsum is coordinatewise dominated by x_F , and the reverse inequality follows by choosing the whole set F .

Choose $f \in S_{X^*}$ with $f(x_F) = \|x_F\|$. Define on the coordinate span

$$g(u_n) = |f(u_n)|.$$

For a finitely supported $y = \sum_n y_n u_n$, choose unimodular scalars θ_n so that $\theta_n f(u_n) = |f(u_n)|$. Then

$$|g(y)| \leq \sum_n |y_n| |f(u_n)| = \left| f \left(\sum_n |y_n| \theta_n u_n \right) \right| \leq \left\| \sum_n |y_n| \theta_n u_n \right\| = \|y\|.$$

Thus g extends to a positive functional in B_{X^*} . Since x_F has nonnegative coordinates,

$$g(x_F) = \sum_{n \in F} c_n |f(u_n)| \geq |f(x_F)| = \|x_F\|,$$

and the opposite inequality follows from $\|g\| \leq 1$. Hence $g(x_F) = \phi(F)$.

Define a finitely supported measure μ_F by

$$\mu_F(\{n\}) = \begin{cases} c_n g(u_n), & n \in F, \\ 0, & n \notin F. \end{cases}$$

For every $B \subseteq \mathbb{N}$,

$$\mu_F(B) = g\left(\sum_{n \in B \cap F} c_n u_n\right) \leq \left\| \sum_{n \in B \cap F} c_n u_n \right\| \leq \phi(B),$$

while $\mu_F(F) = \phi(F)$. Therefore, for arbitrary $A \subseteq \mathbb{N}$,

$$\sup_{\substack{\mu \text{ finitely supported} \\ \mu \leq \phi}} \mu(A) \geq \sup_{F \in [A]^{<\omega}} \mu_F(A) = \sup_{F \in [A]^{<\omega}} \phi(F) = \phi(A).$$

The reverse inequality is automatic from $\mu \leq \phi$. This is exactly (2). $\square$

Remark 3.2. For a merely unconditional basis, an equivalent renorming makes it 1-unconditional, so the same ideal is non-pathological.

3.1 Relation to the established representability theorem

The preceding theorem should not be presented as a new characterization. Borodulin-Nadzieja, Farkas, and Plebanek call an ideal Banach-representable if it has the form

$$\left\{ A \subseteq \mathbb{N} : \sum_{n \in A} x_n \text{ converges unconditionally in a Banach space} \right\}.$$

They proved that the Banach-representable ideals are exactly the non-pathological analytic P -ideals [2]; the later paper [3] develops the connection with Banach spaces having unconditional bases in greater depth. Taking $x_n = c_n u_n$ in Theorem 3.1 gives a particularly simple positive, coordinatewise instance of their result.

Corollary 3.3. *The positive weighted-subseries mechanism used in Theorem 7.1 of [7] cannot realize a pathological analytic P -ideal merely by replacing the Banach sequence norm while retaining an unconditional coordinate basis.*

Proof. In that construction, membership is equivalent to convergence of a positive weighted coordinate subseries. Apply Theorem 3.1. $\square$

Thus any realization of a pathological analytic P -ideal, should one exist, must exploit genuinely different behavior: for example, non-unconditional geometry or the finite-union structure in the definition of a critical ideal.

4 Residual projections and the strong-operator reformulation

Throughout the rest of the note, assume that $\mathcal{CR}(X, \hat{a})$ is proper and write

$$\mathcal{I} = \mathcal{CR}(X, \hat{a}).$$

Biorthogonality gives a rigid algebraic structure.

Lemma 4.1 (Nested projection calculus). *For all $m, n \in \mathbb{N}$,*

$$S_n S_m = S_{\min(n, m)}, \quad R_n R_m = R_{\max(n, m)}.$$

Moreover, X is separable and $\text{span}\{a_n : n \in \mathbb{N}\}$ is dense in X .

Proof. The identities follow by applying the coordinate functionals to the finite sums. For density, fix $x \in X$ and $\varepsilon > 0$. Since $B(x, \varepsilon) \in \mathcal{I}$ and $\mathcal{I}$ is proper, some n lies outside $B(x, \varepsilon)$; hence $\|x - S_n x\| < \varepsilon$, while $S_n x$ belongs to the finite-dimensional span of the a_i . $\square$

Because scaling absorbs the threshold, put

$$B_x = \{n : \|R_n x\| \geq 1\}.$$

Then

$$C \in \mathcal{I} \iff \text{there are } F \in \text{Fin} \text{ and } x_1, \dots, x_r \in X \text{ with } C \subseteq F \cup \bigcup_{j=1}^r B_{x_j}. \quad (5)$$

Proposition 4.2 (Strong-operator accumulation). *For $A \subseteq \mathbb{N}$, the following are equivalent:*

- (i) $A \notin \mathcal{I}$;
- (ii) *for every m , every finite family $x_1, \dots, x_r \in X$, and every $\varepsilon > 0$, there is $n \in A \setminus m$ such that $\|R_n x_j\| < \varepsilon$ for all $j \leq r$;*
- (iii) *0 is an ω -accumulation point of $\{R_n : n \in A\}$ in the strong operator topology.*

Proof. The negation of (ii) says that, for some m , finite family, and $\varepsilon > 0$, $A \setminus m$ is covered by the corresponding bad sets. By scaling the vectors, this is exactly membership in the ideal described by (5). The equivalence of (ii) and (iii) is the definition of strong-operator accumulation. $\square$

The distinction between ordinary closure and ω -accumulation is useful: the latter records that the common good indices may always be chosen arbitrarily far out.

5 An exact statistical selection criterion

For a sequence (T_k) of operators, write

$$T_k \longrightarrow_{\mathcal{I}_d} 0 \quad \text{strongly}$$

if, for every $x \in X$ and $\varepsilon > 0$, the set $\{k : \|T_k x\| \geq \varepsilon\}$ has asymptotic density zero.

Theorem 5.1 (Statistical SOT criterion). *Assume that $\mathcal{I} = \mathcal{CR}(X, \hat{a})$ is an analytic P -ideal. The following are equivalent:*

- (i) $\mathcal{I}$ is non-pathological.
- (ii) *For every $A \notin \mathcal{I}$, there is a map $f : \mathbb{N} \rightarrow A$ such that*
 - (a) $f^{-1}[\{n\}] \in \mathcal{I}_d$ for every $n \in A$;
 - (b) $R_{f(k)} \rightarrow_{\mathcal{I}_d} 0$ strongly.
- (iii) *For every $A \notin \mathcal{I}$, there are finitely supported probability measures (ν_j) on A such that*
 - (a) $\nu_j(\{n\}) \rightarrow 0$ for every fixed $n \in A$;
 - (b) $\nu_j(B(x, \varepsilon)) \rightarrow 0$ for every $x \in X$ and $\varepsilon > 0$.

Proof. Assume (i), and fix $A \notin \mathcal{I}$. By Theorem 2.1, there is a map $f : \mathbb{N} \rightarrow A$ witnessing $\mathcal{I} \restriction A \leq_K \mathcal{I}_d$. Since every singleton belongs to $\mathcal{I} \restriction A$, condition (ii)(a) holds. Every bad set $B(x, \varepsilon) \cap A$ belongs to $\mathcal{I} \restriction A$, so its inverse image under f has density zero. This is (ii)(b).

Assume (ii). Given $A \notin \mathcal{I}$, define the empirical measures

$$\nu_j(C) = \frac{1}{j} |\{k < j : f(k) \in C\}| \quad (j \geq 1).$$

They are finitely supported probabilities on A . Conditions (iii)(a) and (iii)(b) are precisely (ii)(a) and (ii)(b).

Finally, assume (iii). For each $A \notin \mathcal{I}$, let

$$\mathcal{J}_A = \{C \subseteq \mathbb{N} : \nu_j(C) \rightarrow 0\}.$$

The rows $(\nu_j(\{n\}))_n$ form a regular nonnegative matrix: row sums are one and each fixed column tends to zero. Thus $\mathcal{J}_A$ is a regular matrix summability ideal. By (5), every $C \in \mathcal{I}$ is contained in a finite set together with finitely many bad sets. Condition (iii)(a) kills the finite set and (iii)(b) kills the bad sets, so $\mathcal{I} \subseteq \mathcal{J}_A$. On the other hand, $\nu_j(A) = 1$, so $A \notin \mathcal{J}_A$. Consequently,

$$\mathcal{I} = \bigcap_{A \notin \mathcal{I}} \mathcal{J}_A.$$

By the matrix part of Theorem 2.1, $\mathcal{I} \restriction A \leq_K \mathcal{I}_d$ for every $A \notin \mathcal{I}$; the first part of that theorem now gives (i). $\square$

Remark 5.2 (What a complete negative solution would require). By Theorem 5.1, proving that every analytic P -ideal critical ideal is non-pathological is equivalent to producing, on each positive set, a diffuse statistical sampling of the residual projections that converges strongly to zero. This is more precise than merely observing that 0 is in the strong-operator closure: Theorem 4.2 supplies arbitrarily good common indices for each finite family of vectors, whereas Theorem 5.1 asks for one sampling that works for every vector.

6 A bounded-positive-subset criterion

The next criterion is elementary but useful. It isolates one concrete place where an ordinary uniform boundedness argument would finish the problem.

Theorem 6.1 (Local boundedness implies non-pathology). *Assume that $\mathcal{I} = \mathcal{CR}(X, \hat{a})$ is an analytic P -ideal. Suppose that for every $A \notin \mathcal{I}$ there is a set $B \subseteq A$, $B \notin \mathcal{I}$, such that*

$$\sup_{n \in B} \|R_n\| < \infty.$$

Then $\mathcal{I}$ is non-pathological. In fact, for every $A \notin \mathcal{I}$ there is $B \subseteq A$, $B \notin \mathcal{I}$, such that

$$\mathcal{I} \restriction B = \text{Fin} \restriction B.$$

Proof. Fix $A \notin \mathcal{I}$ and such a B , and enumerate B increasingly as $B = \{n_k : k \in \mathbb{N}\}$. Let $M = \sup_{n \in B} \|R_n\|$. On the dense coordinate span, $R_n y = 0$ for all sufficiently large n . Given $x \in X$ and $\varepsilon > 0$, choose a finitely supported y with $\|x - y\| < \varepsilon/M$ (with the evident modification if $M = 0$). For all sufficiently large k ,

$$\|R_{n_k} x\| = \|R_{n_k}(x - y)\| \leq M \|x - y\| < \varepsilon.$$

Thus $R_{n_k} \rightarrow 0$ strongly in the ordinary sense.

If $C \in \mathcal{I}$, write C as in (5). Each bad set B_{x_j} meets B only finitely because $R_{n_k} x_j \rightarrow 0$. Hence $C \cap B$ is finite. Therefore $\mathcal{I} \restriction B = \text{Fin} \restriction B$. The increasing enumeration of B witnesses $\mathcal{I} \restriction A \leq_K \text{Fin} \leq_K \mathcal{I}_d$. Apply Theorem 2.1. $\square$

The hypothesis is stronger than non-pathology, but it can be verified by a filter-dependent uniform boundedness principle.

6.1 A rapid⁺ corollary

A subset of $\mathbb{N}$ is $\mathcal{I}^*$ -stationary precisely when it is $\mathcal{I}$ -positive. A filter $\mathcal{F}$ is *rapid*⁺ if every $\mathcal{F}$ -stationary set has an $\mathcal{F}$ -stationary subset whose increasing enumeration dominates any prescribed increasing function. It is a *weak P^+ -filter* if, for each decreasing sequence $(F_k) \subseteq \mathcal{F}$ and each stationary A , there is a stationary $B \subseteq A$ with $B \subseteq^* F_k$ for every k .

De Bondt and Vernaev proved that a filter is a stationary uniform-boundedness filter for Banach spaces exactly when it is rapid⁺ and weak P^+ [4, Theorem 5.2].

Corollary 6.2 (Rapid⁺ dual filters). *Let $\mathcal{I} = \mathcal{CR}(X, \hat{a})$ be an analytic P -ideal. If the dual filter $\mathcal{I}^*$ is rapid⁺, then $\mathcal{I}$ is non-pathological. More strongly, every $A \notin \mathcal{I}$ contains $B \notin \mathcal{I}$ with $\mathcal{I} \restriction B = \text{Fin} \restriction B$.*

Proof. The dual of a P -ideal is a weak P^+ -filter. Indeed, if (F_k) is a decreasing sequence in $\mathcal{I}^*$, the P -ideal property applied to their complements gives $F \in \mathcal{I}^*$ with $F \subseteq^* F_k$ for all k ; intersecting F with a stationary set preserves stationarity.

Put $\mathcal{F} = \mathcal{I}^*$. Since $R_n x \rightarrow_{\mathcal{F}} 0$ for every $x \in X$, the sequence (R_n) is pointwise $\mathcal{F}$ -bounded. By the stationary uniform boundedness principle, for every $\mathcal{F}$ -stationary A there is an $\mathcal{F}$ -stationary $B \subseteq A$ on which the operators R_n are equicontinuous. In normed spaces this means $\sup_{n \in B} \|R_n\| < \infty$. Apply Theorem 6.1. $\square$

This corollary is conditional: it does not show that every dual filter arising from an analytic P -ideal is rapid⁺. It does, however, completely settle the source question for the subclass whose dual filters have that property.

7 A necessary reciprocal-norm condition

A recent theorem of Kadets and Manskova gives a different, quantitative restriction. In their terminology, if (a_n) is an $\mathcal{F}$ -basis with continuous coordinate functionals in an infinite-dimensional Banach space, then the sequence of partial-sum norms is $(\mathcal{F}, 1)$ -admissible [6, Theorem 4.3]. Translating their definition gives the following.

Proposition 7.1 (Reciprocal-norm divergence). *Assume that X is infinite-dimensional and $\mathcal{I} = \mathcal{CR}(X, \hat{a})$. Then for every $A \notin \mathcal{I}$,*

$$\sum_{n \in A} \frac{1}{\|S_n\|} = \infty.$$

Consequently, also

$$\sum_{n \in A} \frac{1}{1 + \|R_n\|} = \infty.$$

In particular, the summable ideal

$$\mathcal{I}_w = \left\{ A \subseteq \mathbb{N} : \sum_{n \in A} \frac{1}{1 + \|R_n\|} < \infty \right\}$$

is contained in $\mathcal{I}$.

Proof. Use the dual filter $\mathcal{F} = \mathcal{I}^*$. Its stationary sets are exactly the sets outside $\mathcal{I}$, so the first assertion is precisely $(\mathcal{F}, 1)$ -admissibility. Since S_n and $R_n = I - S_n$ are nonzero projections in an infinite-dimensional space, their norms are at least one, and

$$\|R_n\| \leq 1 + \|S_n\|.$$

Hence

$$\frac{1}{1 + \|R_n\|} \geq \frac{1}{2 + \|S_n\|} \geq \frac{1}{3\|S_n\|}.$$

This proves the second divergence statement. If $A \in \mathcal{I}_w$ but $A \notin \mathcal{I}$, the second statement gives a contradiction; thus $\mathcal{I}_w \subseteq \mathcal{I}$. $\square$

This is a genuine restriction on every critical ideal, but it does not by itself yield the probability measures in Theorem 5.1. Divergence of reciprocal norms allows the norms to grow without bound on every positive set, and that is precisely the regime in which extension from a dense set of vectors becomes difficult.

8 The remaining selection problem

The P -ideal property already gives a useful preliminary diagonalization. Let (x_j) be dense in X . All sets

$$\{n : \|R_n x_j\| \geq 1/q\} \quad (j, q \geq 1)$$

belong to $\mathcal{I}$. If $A \notin \mathcal{I}$, the P -ideal property gives $C \in \mathcal{I}$ that almost contains all of them. Therefore, on the positive set $A \setminus C$, $R_n x_j \rightarrow 0$ ordinarily for every j . What is missing is the passage from the dense set (x_j) to all of X when $\|R_n\|$ is unbounded.

Selection problem 8.1 (Statistical strong-operator selection). *Let (R_n) be the residual projections of an ideal Schauder system, and suppose $\mathcal{I} = \mathcal{CR}(X, \hat{a})$ is an analytic P -ideal. For every $A \notin \mathcal{I}$, must there exist a map $f : \mathbb{N} \rightarrow A$ with density-zero fibers such that*

$$R_{f(k)} \longrightarrow_{\mathcal{I}_d} 0 \quad \text{strongly?}$$

Equivalently, must there be diffuse finitely supported probabilities on A for which $R_n x \rightarrow 0$ in probability for every $x \in X$?

By Theorem 5.1, an affirmative answer to Theorem 8.1 would prove that every analytic P -ideal critical ideal is non-pathological. Since pathological analytic P -ideals are known to exist, that would give a negative answer to Question 11.2 of [7]. Conversely, a realization of a pathological analytic P -ideal would necessarily furnish a counterexample to the selection principle.

The algebra

$$R_n R_m = R_{\max(n, m)}$$

is extra structure not present in an arbitrary pointwise-null sequence of operators, and it remains a plausible source of further progress. The results above show exactly what that structure would have to provide: either a bounded positive thinning, or at least a diffuse statistical thinning with strong convergence.

9 Literature relation and final assessment

The relationship among the results can be summarized as follows.

1. Kwela–Swaczyna realize every non-pathological analytic P -ideal as a critical ideal by a simple construction over an unconditional coordinate basis [7, Theorem 7.1].
2. The packet’s unconditional-support theorem is correct after the repairs in Theorem 3.1, but it is subsumed by the Borodulin-Nadzieja–Farkas–Plebanek characterization of Banach-representable ideals [2].
3. The Hrušák/Filipów–Tryba criterion translates the unresolved converse into the exact statistical SOT condition of Theorem 5.1 [5].
4. De Bondt–Vernaev’s filter-dependent uniform boundedness theorem gives the rapid⁺ case, Theorem 6.2 [4].
5. Kadets–Manskova’s theorem gives the reciprocal-norm obstruction in Theorem 7.1 [6].
6. Avilés–Rosendal–Taylor–Tradacete prove that every basis with respect to an analytic filter is also a basis with respect to some Borel filter [1]. This gives a Borel envelope but does not identify the critical ideal and does not resolve Question 11.2.

Status statement 9.1. The full realization question remains unresolved by the present argument. The best complete statements obtained here are:

- the corrected unconditional-support obstruction;
- the exact statistical-SOT and matrix-averaging characterization;
- non-pathology under bounded positive thinning, in particular under a rapid⁺ dual filter;
- reciprocal-norm divergence on every positive set.

No claim of novelty is made for the unconditional theorem, and the operator-theoretic reductions should be independently checked before being cited as new results.

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